Genomic insights into *Aeluropus lagopoides a*daptation strategies for extreme environment

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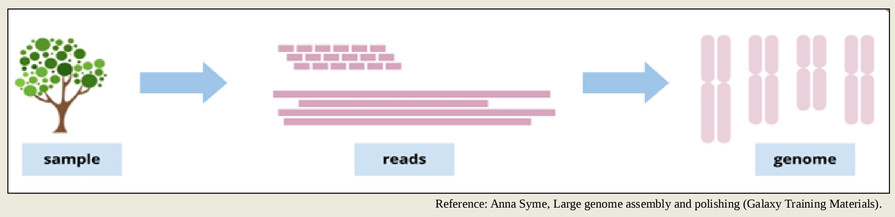
Introduction:



Aeluropus lagopoides or rabbit-foot aeluropus is a highly salt-tolerant mangrove grass of family Poaceae that thrives in extreme saline environments. Understanding its genome provides valuable insights into genetic signatures governing molecular interactions that impart salt tolerance. Further dissection of which has immense implication in potentiating resilience in commercially relevant crops.

Genome assembly is crucial for identification of genes responsible for ion homeostasis, osmoprotectant synthesis and salt sequestration mechanisms. A well-annotated genome enhances our understanding of how plants adapt to high-salinity conditions, paving the way for agricultural innovations in saline-affected regions.

This study is a collaborative effort between CSIR-CSMCRI and CSIR-CCMB, utilizing advanced sequencing technologies to construct a high-quality genome assembly.



Reference: Anna Syme, Large genome assembly and polishing (Galaxy Training Materials).

Figure 1: **Simplified diagram showing workflow of plant genome assembly**

Objective:

- ✓ **Sequence and assemble** the complete genome of *Aeluropus lagopoides*
- ✓ **Compare** the genomic features of *Aeluropus lagopoides* with other halophytic species to identify **unique** adaptive traits.
- ✓ A**nnotate** the genome to identify genes associated with **salt and drought tolerance**.

Methods:

- ❑ A hybrid genome assembly approach was employed using both long-read and short-read sequencing technologies. **PacBio HiFi sequencing** provided high-contiguity reads, while **Illumina NovaSeq 6000** short reads improved accuracy.
- ❑ To enhance genome scaffolding, **Optical Genome Mapping (OGM)** was performed using Bionano technology. Genome assembly was conducted using **Masurca** for hybrid assembly, **Flye** for long-read assembly, and **Pilon** for polishing.
- ❑ Gene annotation was performed using Augustus and BRAKER, focusing on salt tolerance-related genes.
- ❑ Quality control assessments, including BUSCO, QUAST, and FastQC, ensured high completeness and accuracy of the final assembly.

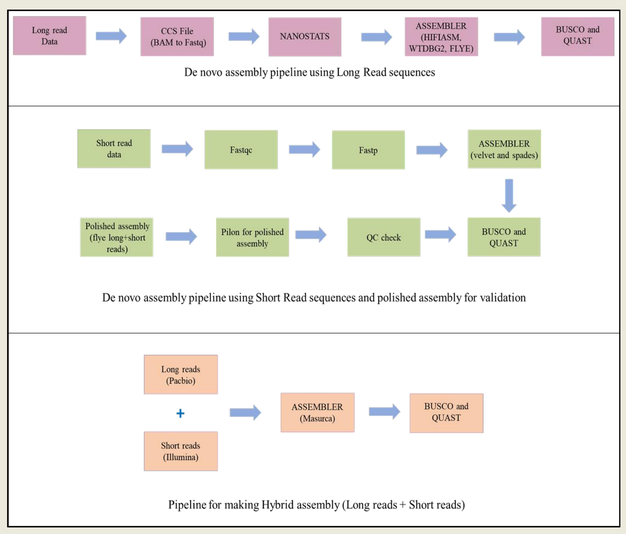


***Figure 2*: A hybrid approach to *de-novo* genome assembly.**

Results:

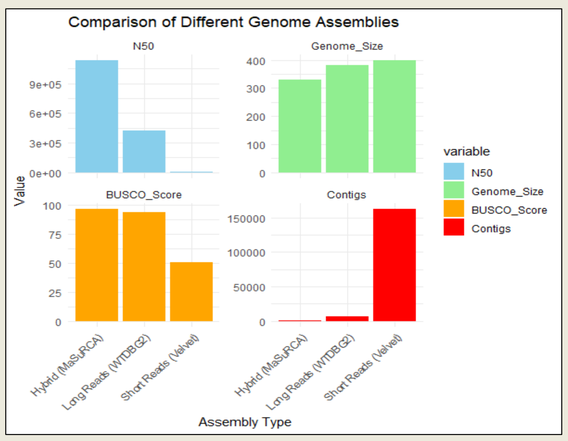


Figure 3: **Comparison between different genome assembly methods based on key quality metrics:**

N50 (blue): Measures assembly contiguity; higher values indicate better assemblies.
Genome Size (green): Estimated genome size in megabases (Mb) for each assembly approach.
BUSCO Score (orange): Assesses genome completeness based on conserved orthologs; higher scores suggest better assemblies.
Number of Contigs (red): Represents genome fragmentation; fewer contigs indicate a more contiguous assembly.
Three assembly strategies are evaluated:
Hybrid (MaSuRCA): Combines short and long reads for improved accuracy.
Long Reads (WTDBG2): Uses only long-read sequencing for assembly.
Short Reads (Velvet): Uses only short-read sequencing, resulting in highly fragmented assemblies.

This comparison highlights that **hybrid assembly** provides the best balance between contiguity, completeness, and genome size accuracy.

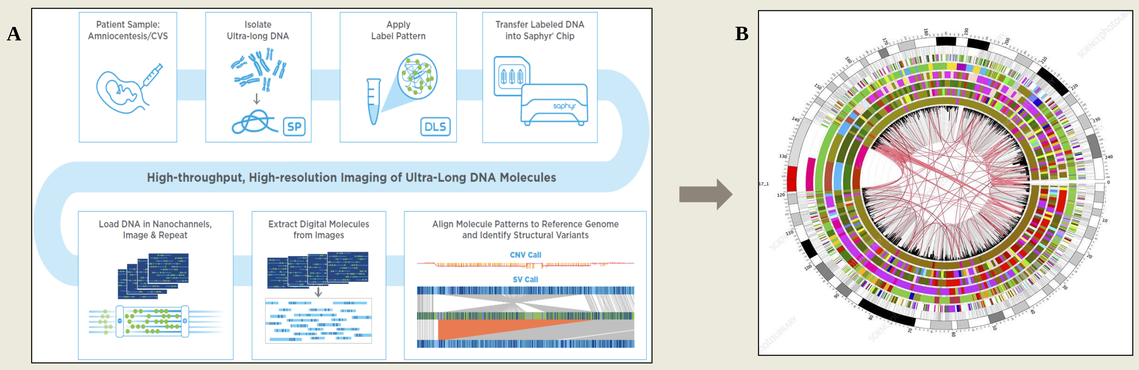
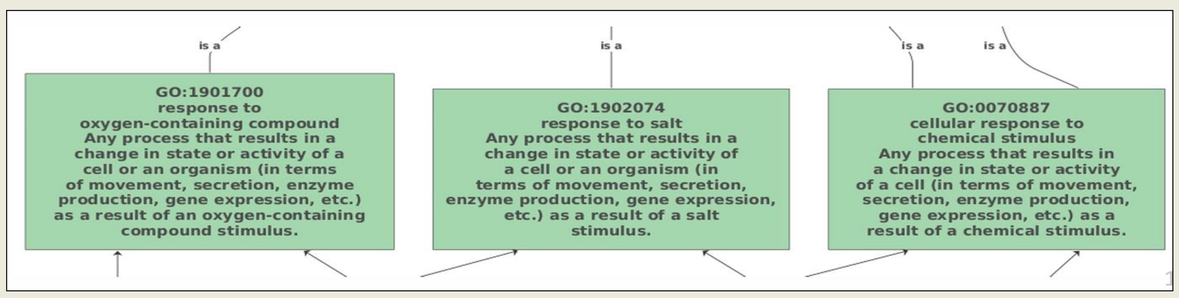


Figure 4:**A and B.** A representation of OGM (Optical Genome Mapping) that outputs a ultra long structure of complete genome architecture .

C. BNX format file generated in this process contains all the GO annotations which can be processed through official bionano server and then annotation through Omics box gene ontology. This provides all the chromosomal level information.

Here, we show gene ontology result which sheds light on the sets of genes that might be relevant to the salt tolerance mechanisms in ***Aeluropus lagopaoides.***



Discussion:

The **hybrid genome assembly** approach significantly improved genome completeness and structural accuracy compared to individual sequencing methods. **Bionano Optical Mapping** played a key role in scaffolding improvements, enhancing genome contiguity. The high **BUSCO completeness scores** indicate a reliable and well-annotated genome. Functional analysis identified critical **genes related to salt stress tolerance, ion transport, and osmoprotection**, confirming *Aeluropus lagopoides* as an ideal model for studying saline adaptation.
Future work will focus on **validating gene functions** through experimental approaches and utilizing this genomic resource for **crop improvement in saline-affected regions.**

Conclusion:

- A high-quality genome of *Aeluropus lagopoides* was assembled using a hybrid approach.
- Hybrid assembly improved genome completeness and gene annotation.
- Key salt tolerance genes related to ion homeostasis and osmoprotection were identified.
- Optical Genome Mapping (OGM) enhanced genome structural accuracy.
- This study provides a valuable resource for future research and crop improvement.

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References:

1. Sahajpal, N. S., Barseghyan, H., Kolhe, R., Hastie, A., & Chaubey, A. (2021). Optical Genome Mapping as a Next-Generation Cytogenomic Tool for Detection of Structural and Copy Number Variations for Prenatal Genomic Analyses. *Genes*, *12*(3), 398. https://doi.org/10.3390/genes12030398
2. Hashemi-Petroudi, S. H., Arab, M., Dolatabadi, B., Kuo, Y. T., Baez, M. A., Himmelbach, A., Nematzadeh, G., Maibody, S. A. M. M., Schmutzer, T., Mälzer, M., Altmann, T., & Kuhlmann, M. (2022). Initial Description of the Genome of *Aeluropus littoralis*, a Halophile Grass. *Frontiers in plant science*, *13*, 906462. https://doi.org/10.3389/fpls.2022.906462
3. Younesi-Melerdi, E., Nematzadeh, GA., Pakdin-Parizi, A. *et al. De novo* RNA sequencing analysis of *Aeluropus littoralis* halophyte plant under salinity stress. *Sci Rep* **10**, 9148 (2020). https://doi.org/10.1038/s41598-020-65947-5

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